



**Hanoi School
of Business &
Management**

TRƯỜNG QUẢN TRỊ VÀ KINH DOANH
ĐẠI HỌC QUỐC GIA HÀ NỘI - VNU



HSB1003 Data Analysis

MIDTERM EXAM

Undergraduate students

Lecturers:

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Time: 90 mins

PART A: 20 points

Choose the best answer (A, B, C, or D) for each question.

1. A survey asks consumers to rate their agreement with the statement “This brand is trustworthy” on a 1–5 Likert scale (Strongly disagree → Strongly agree). What is the measurement scale?
A. Nominal
B. Ordinal
C. Interval
D. Ratio
2. A company evaluates customer satisfaction by surveying only customers who follow the brand’s Instagram account. What issue is most likely?
A. Random sampling error
B. Selection bias (non-representative sample)
C. Measurement bias caused by a faulty scale
D. Confounding due to a third variable
3. A dataset of monthly customer spending is highly right-skewed with several extreme outliers. Which summary is most appropriate?
A. Mean and standard deviation
B. Median and interquartile range (IQR)
C. Mode and range
D. Correlation and covariance
4. Which statement about the correlation coefficient (r) is TRUE?

- A. r close to 0 means there is no relationship of any kind
- B. r can only be positive
- C. r measures the strength and direction of a linear relationship
- D. r is not affected by changing units (e.g., dollars to cents)
5. If $P(A) = 0.30$, what is the probability of the complement event A^c (not A)?
- A. 0.30
- B. 0.70
- C. 1.30
- D. Cannot be determined
6. A class has 10 students: 4 females and 6 males. If one student is selected randomly, what is the probability that the student is female?
- A. 0.30
- B. 0.40
- C. 0.50
- D. 0.60
7. In a city, 30% of days are rainy. When it is rainy, flooding occurs 40% of the time. What is the probability that a randomly selected day is both rainy and flooded?
- A. 0.12
- B. 0.70
- C. 0.40
- D. 0.30
8. If events A and B are independent, which formula is correct?
- A. $P(A \cap B) = P(A) + P(B)$
- B. $P(A \cap B) = P(A)P(B)$
- C. $P(A|B) = P(B|A)$
- D. $P(A \cup B) = P(A)P(B)$
9. For a normal distribution, approximately 95% of observations lie within which interval?
- A. $\mu \pm 1\sigma$
- B. $\mu \pm 1.645\sigma$
- C. $\mu \pm 1.96\sigma$
- D. $\mu \pm 2.576\sigma$
10. You want a confidence interval for a population mean μ when the population standard deviation σ is unknown, the population is approximately normal, and n is small (e.g., $n = 15$). Which distribution should be used?
- A. Standard normal (z)

B. Chi-square

C. Student's t with $df = n - 1$

D. F distribution

PART B:

Exercise 1: 30 points

See the excel file

Exercise 2: 30 points

A computer software supplier has developed a new record-keeping package for use by hospitals. The company feels that the probability that the new package will show a profit in its first year is 0.6, unless a competitor introduces a product of comparable quality this year, in which case the probability of a first-year profit drop to 0.3. The supplier suggests that there is a 50/50 chance that a comparable product will be introduced this year. We define the following events:

A: A competitor introduces a comparable product.

B: the record-keeping package is profitable in its first year.

Calculate the following probabilities:

- a) What is the probability that both A and B will occur?

Xác suất cả A và B xảy ra

$$P(A \cap B) = P(A) \times P(B|A) = 0.5 \times 0.3 = 0.15$$

- b) What is the probability that either A or B will occur?

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) = 0.3 \times 0.5 + 0.6 \times 0.5 = 0.15 + 0.30 = 0.45$$

$$\Rightarrow P(A \cup B) = 0.5 + 0.45 - 0.15 = 0.80$$

- c) What is the probability that neither A nor B will occur?

$$P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.8 = 0.2$$

Exercise 3: 20 points

Students investigating the packaging of potato chips purchased 6 bags of Lay's Ruffles marked with a net weight of 28.3 grams. They carefully weighed the contents of each bag, recording the following weights (in grams): 29.3, 28.2, 29.1, 28.7, 28.9, 28.5

- (a) Find the mean and standard deviation of the observed weights. (Please calculate in Excel)

- (b) Create a 95% confidence interval for the mean weight of such bags of chips.

- (c) Explain in context what your interval means.